

$$\text{II} \quad \frac{1}{2} m V_a^2 + mgL = \frac{1}{2} m V_b^2$$

$$V_a^2 - V_b^2 = -2gL$$

$$V_a^2 = V_b^2 - 2gL$$

$$= 25 - 2 \cdot 10 \cdot 1$$

$$= 5$$

$$V_a = \sqrt{5} \text{ (m/s)}$$

$$m \cdot \frac{V_a^2}{r} = mg + T_a$$



$$2 \cdot \frac{5}{1} = 20 + T_a$$

$$T_a = -10 \text{ (N)}$$

$$\frac{1}{2} m V_b^2 + mgL = \frac{1}{2} m V_c^2$$

$$V_c^2 = V_b^2 + 2gL$$

$$= 25 + 2 \cdot 10 \cdot 1$$

$$= 45$$

$$V_c = 3\sqrt{5} \text{ (m/s)}$$

$$m \cdot \frac{V_c^2}{r} = T_c - mg$$

$$T_c = mg + m \cdot \frac{V_c^2}{r}$$

$$= 2 \cdot 10 + 2 \cdot \frac{45}{1}$$

$$= 110 \text{ (N)}$$

· 장력이 끊어 해준일

$$W_g = \frac{1}{2} m V_c^2 - \frac{1}{2} m V_a^2$$

$$= \frac{1}{2} \cdot 2 \cdot 45 - \frac{1}{2} \cdot 2 \cdot 5$$

$$= 40 \text{ (J)}$$

· 장력이 끊어 해준일

$$W_T = 0 \quad (\because \text{radial 방향으로 } \text{한번도 없음})$$